Ch 3.3: The Last of the Linear Regression

Lecture 8 - CMSE 381

Prof. Elizabeth Munch

Michigan State University
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Dept of Computational Mathematics, Science & Engineering

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Announcements

Last time:

- Started 3.3 Questions of linear regression

Announcements:

- HW #~~4~~ 3 Due Sunday 9/22

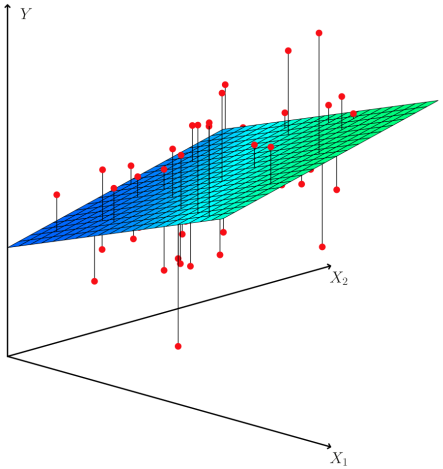
Covered in this lecture

- Extending the linear model with interaction terms
- Hierarchy principle
- Polynomial regression

Section 1

Review from last time

Linear Regression with Multiple Variables



- Predict Y on a multiple variables X

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots \beta_p x_p + \varepsilon$$

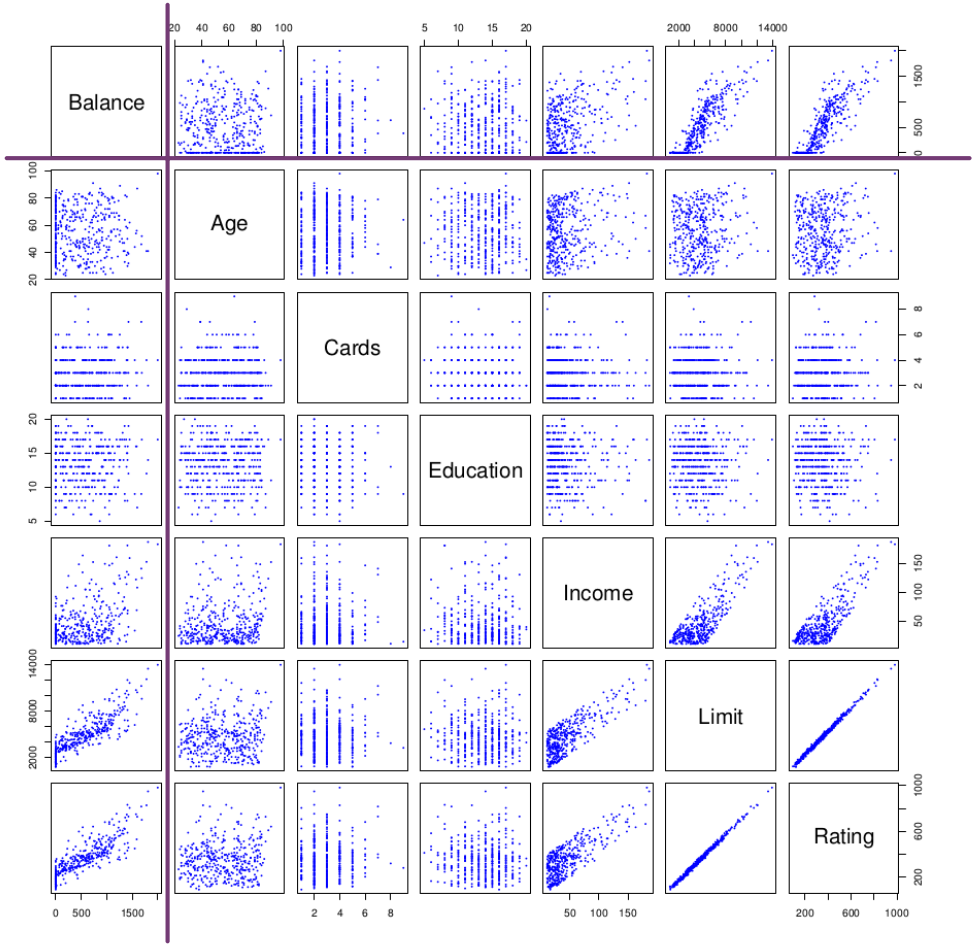
- Find good guesses for $\hat{\beta}_0$, $\hat{\beta}_1, \cdots$.
- $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \cdots + \hat{\beta}_p x_p$

- $e_i = y_i - \hat{y}_i$ is the ith residual
- RSS $= \sum_i e_i^2$
- RSS is minimized at *least squares coefficient estimates*

Section 2

Categorical Input Variables

Credit card balance



quantitative input variables

qualitative input variables

- `own`: house ownership yes/no
- `student`: student status yes/no
- `status`: marital status yes/no
- `region`: East, West, or South

3 levels

One-hot encoding of categorical input variable

Ex) the student variable

called a "dummy variable"

Create a new variable

$$x_i = \begin{cases} 1 & \text{if } i\text{th person is a student} \\ 0 & \text{if } i\text{th person is not a student} \end{cases}$$

Model:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$= \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{th person is student} \\ \beta_0 + \varepsilon_i & \text{if } i\text{th person isn't} \end{cases}$$

Who cares about 0/1?

Old version: 0/1

$$x_i = \begin{cases} 1 & \text{if } i\text{th person is a student} \\ 0 & \text{if } i\text{th person is not a student} \end{cases}$$

Model:

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_i + \varepsilon_i \\ &= \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{th person is student} \\ \beta_0 + \varepsilon_i & \text{if } i\text{th person isn't} \end{cases} \end{aligned}$$

Alternative version: ± 1

$$x_i = \begin{cases} 1 & \text{if } i\text{th person is a student} \\ -1 & \text{if } i\text{th person is not a student} \end{cases}$$

Model:

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_i + \varepsilon_i \\ &= \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{th person is student} \\ \beta_0 - \beta_1 + \varepsilon_i & \text{if } i\text{th person isn't} \end{cases} \end{aligned}$$

Qualitiative Predictor with More than Two Levels

↳ # of values the variable can take

Region:

South?

	x_{i1}	x_{i2}	x_{i3}
South	1	0	0
West	0	1	0
East	0	0	1

Levels = 3

Create spare dummy variables:

$$x_{i1} = \begin{cases} 1 & \text{if } i\text{th person from South} \\ 0 & \text{if } i\text{th person not from South} \end{cases}$$

East or West

$$x_{i2} = \begin{cases} 1 & \text{if } i\text{th person from West} \\ 0 & \text{if } i\text{th person not from West} \end{cases}$$

unnecessary b/c can be determined from the other two.

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$$

$$= \begin{cases} \beta_0 + \beta_1 \cancel{x_{i1}} + \varepsilon_i & \text{if } i\text{th person from South} \\ \beta_0 + \beta_2 \cancel{x_{i2}} + \varepsilon_i & \text{if } i\text{th person from West} \\ \beta_0 + \varepsilon_i & \text{if } i\text{th person from East} \end{cases}$$

More on multiple levels

- An input variable with k levels, use k-1 dummy variables
- Level with no dummy variable is called the $\boxed{\text{baseline}}$

	Coefficient	Std. error	t-statistic	p-value
Intercept	$\beta_0 = 531.00$	46.32	11.464	< 0.0001
region[South]	$\beta_1 = -18.69$	65.02	-0.287	0.7740
region[West]	$\beta_2 = -12.50$	56.68	-0.221	0.8260

In this example: baseline is east.

East $\rightarrow$ always predict β_0
\$531

South $\rightarrow$ always predict $\beta_0 + \beta_1$
$531 - 18.69$

West $\rightarrow$ always predict $\beta_0 + \beta_2$
$531 - 12.5$

High p-values for dummy variables
No statistical evidence that regions have diff. balance

Do code section on "Dummy Variables for Multi-level Categorical Inputs"

Section 3

Extending the linear model

Assumptions so far

Back to our Advertising data set

$$\hat{Y}_{sales} = \beta_0 + \beta_1 \cdot X_{TV} + \beta_2 \cdot X_{radio} + \beta_3 \cdot X_{newspaper}$$

Assumed (implicitly) that the effect on sales by increasing one medium is independent of the others.

What if spending money on radio advertising increases the effectiveness of TV advertising? How do we model it?

Interaction Term

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \underbrace{\beta_3 X_1 X_2}_{\text{interaction term}} + \varepsilon$$

$$\begin{aligned} Y_{sales} &= \beta_0 + \beta_1 X_{TV} + \beta_2 X_{radio} + \beta_3 X_{radio} X_{TV} + \varepsilon \\ &= \beta_0 + (\beta_1 + \beta_3 X_{radio}) X_{TV} + \beta_2 X_{radio} + \varepsilon \\ &= \beta_0 + \tilde{\beta}_1 X_{TV} + \beta_2 X_{radio} + \varepsilon \end{aligned}$$

$\tilde{\beta}_1 = \beta_1 + \beta_3 X_{radio}$

function of X_{radio}

no longer constant

b/t X_{TV} and Y

Interaction term

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 \boxed{X_1 X_2} + \varepsilon$$

	Coefficient	Std. error	t-statistic	p-value
Intercept	6.7502	0.248	27.23	< 0.0001
TV	$\beta_1 =$ 0.0191	0.002	12.70	< 0.0001
radio	$\beta_2 =$ 0.0289	0.009	3.24	0.0014
TV×radio	$\beta_3 =$ 0.0011	0.000	20.73	< 0.0001

$$\begin{aligned} Y_{sales} &= \beta_0 + \beta_1 X_{TV} + \beta_2 X_{radio} + \beta_3 X_{radio} X_{TV} + \varepsilon \\ &= \beta_0 + (\beta_1 + \beta_3 X_{radio}) X_{TV} + \beta_2 X_{radio} + \varepsilon \end{aligned}$$

Small p-value for interaction term
$\Rightarrow$ Strong evidence for $\beta_3 \neq 0$

Interpretation: β_3 the increase in effectiveness of TV advertising associated with 1 unit increase in radio

Interpretation

	Coefficient	Std. error	t-statistic	p-value
Intercept	6.7502	0.248	27.23	< 0.0001
TV	0.0191	0.002	12.70	< 0.0001
radio	0.0289	0.009	3.24	0.0014
TV×radio	0.0011	0.000	20.73	< 0.0001

- Increase in \$1K of TV is associated to increased sales by

$$(\beta_1 + \beta_3 X_{radio})(\text{change in } X_{TV})$$

$$(0.0191 + 0.0011 X_{radio})(1000)$$

$$19 + 1.1 \times X_{radio}$$

Can do the same for increase in radio:

$$(\beta_2 + \beta_3 X_{TV}) \cdot (\text{change in } X_{radio})$$

Hierarchy principle

Sometimes p-value for interaction term is very small, but associated main effects are not.

Bad: $Y \approx \beta_0 + \beta_1 X_{TV} X_{radio} + \varepsilon_i$

Good: $Y \approx \beta_0 + \beta_1 X_{TV} + \beta_2 X_{radio} + \beta_3 X_{TV} X_{radio} + \varepsilon$

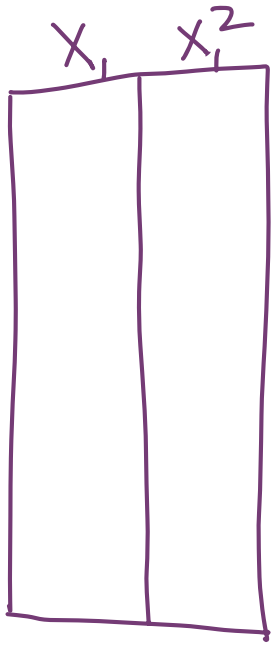
The hierarchy principle:

If we include the interaction term, we should always include the main terms

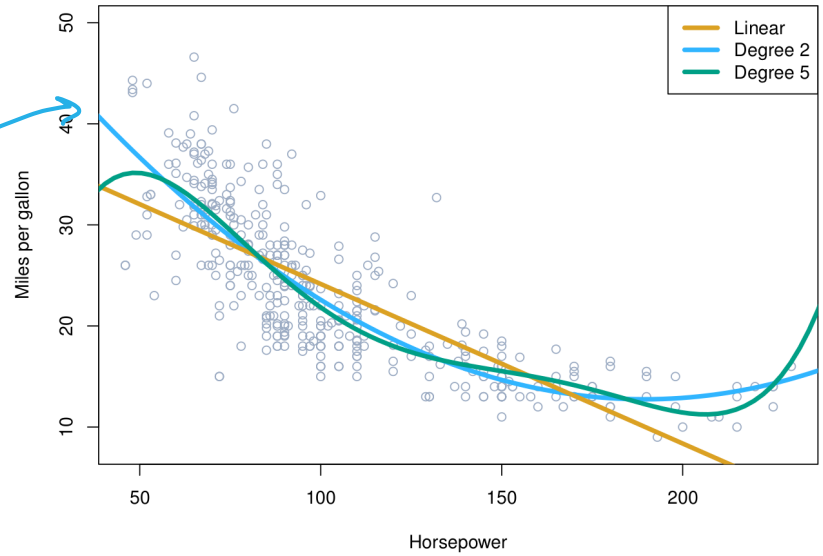
Nonlinear relationships

$$\texttt{mpg} = \beta_0 + \beta_1 \cdot \texttt{horsepower} + \beta_2 \cdot \texttt{horsepower}^2 + \varepsilon$$

Still done w/ lin reg code



degree 2 model



	Coefficient	Std. error	t-statistic	p-value
`Intercept`	56.9001	1.8004	31.6	< 0.0001
`horsepower`	-0.4662	0.0311	-15.0	< 0.0001
`horsepower`^{2}	0.0012	0.0001	10.1	< 0.0001

Do the section on “Interaction Terms”

Next time

Lec #	Date			Reading	HW
1	Mon	8/26	Intro / First day stuff / Python Review Pt 1	1	
2	Wed	8/28	What is statistical learning?	2.1	
	Fri	8/30	Class Cancelled (Dr Munch out of town)		
	Mon	9/2	No class - Labor day		
3	Wed	9/4	Assessing Model Accuracy	2.2.1, 2.2.2	
4	Fri	9/6	Linear Regression	3.1	HW #1 Due Sun 9/8
5	Mon	9/9	More Linear Regression	3.1/3.2	
6	Wed	9/11	Even more linear regression	3.2.2	
7	Fri	9/13	Probably more linear regression	3.3	Hw #2 Due Dun 9/15
8	Mon	9/16	Linear regression coding module		
9	Wed	9/18	Intro to classification, Bayes classifier, KNN classifier	2.2.3	
10	Fri	9/20	Logistic Regression	4.1, 4.2, 4.3.1-3	Hw #3 Due Sun 9/22
11	Mon	9/23	Multiple Logistic Regression / Multinomial Logistic Regression /Project day	4.3.4-5	
	Wed	9/25	***Review***		